

$$f_u = \frac{g_m}{2\pi C_{out,tot}}$$

$$GBW_{OL} = GBW_{CL}$$

$$\frac{C_{in}}{C_{in} + C_F} \left[A_o BW_{oIA} \right] = A_{CL} BW_{CL}$$

$$\rightarrow B = \frac{C_F}{C_F + C_S} = \frac{1}{1+2} = \frac{1}{3} \rightarrow A_{CL} = \frac{1}{B} = 3$$

$$GBW_{OTA} = \frac{3 \times 10^6 BW_{CL}}{3}$$

$$(GBW_{OTA} = 3 BW_{CL})$$

$$\rightarrow \rho = 0.01 \quad t = 100ns \quad 1 = 21.7ns$$

$$f_{CL} = 7.33 MHz$$

$$\rightarrow GBW = A_{CL} \times BW_{CL} = \frac{BW_{CL}}{3} = 22.1 MHz$$

$$\rightarrow C_{out,tot} = C_L + (C_F \parallel \frac{C_S}{C_{in}}) = 1.167 pF$$

$$\therefore \text{So } g_{m1,2} = 2\pi GBW_{CL} C_{out,tot} \approx 161.3 \mu$$

$$\rightarrow G_m = g_{m1,2}$$

$$R_{out} = (r_{o3} \parallel g_{m3} \parallel r_{o2} \parallel r_{o1}) \parallel r_{o4} \parallel g_{m4} \parallel r_{o5}$$

$$V_{max} = 100 - V_{GS1} - V_{DSAT6}$$

$$\textcircled{1} g_m = 165 \text{ m}$$

$$2.5 - V_{GS1} - V_{DSAT6} > 1$$

$$1.5 > V_{GS1} + V_{DSAT6}$$

$$LG = 1000 = \beta A_{OL}$$

$$A_{OL} = 3000$$

$$\text{let } g_{m/10} = 16.5 \rightarrow I_D = 10 \mu\text{A}$$

$$V_{GS1} = 7.82 \text{ V} \quad W = 20.88$$

$$V_{DSAT6} < 0.7176$$

$$V_{GS1} \approx 1.125 \quad V_{GS2} \approx 1.244$$

$$3.96 \text{ m}$$

$$L = 1 \mu\text{m}$$

$$1.5 \text{ m}$$